



A decision support methodology for strategic planning in maritime transportation

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ABSTRACT

This paper presents a decision support methodology for strategic planning in tramp and industrial shipping. The proposed methodology combines simulation and optimization, where a Monte Carlo simulation framework is built around an optimization-based decision support system for short-term routing and scheduling. The simulation proceeds by considering a series of short-term routing and scheduling problems using a rolling horizon principle where information is revealed as time goes by. The approach is flexible in the sense that it can easily be configured to provide decision support for a wide range of strategic planning problems, such as fleet size and mix problems, analysis of long-term contracts and contract terms. The methodology is tested on a real case for a major Norwegian shipping company. The methodology provided valuable decision support on important strategic planning problems for the shipping company.

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1. Introduction

The world economy is heavily dependent on maritime transportation, which is probably the most important mode of transportation. More than 7 billion tonnes annually are carried by ships, and this figure seems to increase every year [1]. A ship involves a major capital investment (usually millions of US dollars, tens of millions for larger ships) and a large long-term transport contract for a ship operator may span several years and require a considerable increase in fleet size. Proper planning of fleets and their operations has the potential of improving their economic performance and reducing shipping costs. Here we focus on the strategic decisions for a ship operator, ranging from fleet size and mix decisions to transport contract considerations. To ensure good long-term, strategic decisions we also consider a significant amount of details regarding short-term decisions.

Usually we distinguish between three general modes of shipping operations: *Industrial*, *tramp*, and *liner shipping* [2]. In industrial shipping, the cargo owner or shipper also controls the fleet of ships. Industrial operators try to ship all their cargoes at minimum cost. Tramp shipping is similar to a taxi operation, where the ships follow the available cargoes. A tramp shipping company often has a certain amount of contract cargoes that can be considered as mandatory. In addition, the company tries to

maximize profit from optional spot cargoes. Liners operate similarly to a bus line, with published itineraries and schedules. The distinction between these modes is not always clear, and shipping companies may simultaneously operate their fleets in different modes. In this paper we limit ourselves to industrial and tramp shipping. Typical cargoes carried in these modes of operation include various dry and wet bulk products. For these two modes, there are a few references on routing and scheduling, see for instance [3–7] and [8]. However, all these studies involve short-term routing and scheduling, which can be considered as tactical/operational planning problems.

The literature is scarce considering studies of strategic planning problems in maritime transportation. We can find some contributions dealing with optimal fleet size and mix problems, see for instance [9] and [10]. However, most of the studies dealing with strategic maritime transportation problems are for liner shipping (including [9] and [10]). The few contributions that are also valid for industrial/tramp are developed to solve only very specialized problems. One exception here is [11], where a model is developed for evaluating the effectiveness of alternative policies regarding chartering out vessels on a long-term basis or operating in the spot/contract market.

For a review on the use of operational research (OR) in maritime transportation, we refer to [12,13] and [14]. Furthermore, in [15], a comprehensive survey of various planning problems in maritime transportation is given, together with a presentation of mathematical models and a discussion of solution methods for some of the problems.

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The objective of this paper is to present a new decision support methodology that is applicable to a wide range of strategic planning problems in industrial and tramp shipping. The methodology includes a Monte Carlo simulation framework built around an optimization-based tool for solving the underlying short-term routing and scheduling problem. The methodology can provide support for ship operators in decision problems related to for instance fleet size and mix and the evaluation of transport contracts.

The rest of the paper is organized as follows. Section 2 gives a thorough description of these focused strategic planning problems and of how they usually are or can be solved, both in practice and in the research literature. Section 3 presents the proposed decision support methodology for these problems. A case study where we have applied the decision support methodology to provide decision support for an industrial shipping company in a real setting is described in Section 4. Concluding remarks are given in Section 5.

2. Strategic planning in industrial and tramp shipping

Strategic planning in industrial and tramp shipping spans a large variety of problems. Since there is a strong interplay between strategic, tactical and operational planning levels, one often cannot provide valuable decision support for strategic problems without simultaneously considering the underlying routing decisions. This is also emphasised in the first review paper on ship routing and scheduling [14]. Therefore, we start in Section 2.1 by briefly describing the underlying short-term routing and scheduling problem in industrial and tramp shipping. Section 2.2 presents the most important strategic planning decisions on which we focus in this paper, while a discussion of various solution approaches is given in Section 2.3.

2.1. The short-term routing and scheduling problem

The short-term routing and scheduling problem in industrial and tramp shipping consists of optimally assigning cargoes to available ships in the fleet and simultaneously deciding the underlying routes and schedules for each ship. Each cargo consists of a pickup port, a delivery port, a quantity of a given product (can be given in an interval), and a time window (for loading and/or discharging) in which the loading/discharging must start. To service the cargoes, the ship operator controls a fleet of ships, each ship with a given capacity and service speed. In contrast to traditional Vehicle Routing Problem the ships do not return to a depot, and their initial and ending positions in a given planning problem can be any port or point at sea. The ship routing and scheduling problem is similar to a multi-vehicle Pickup and Delivery Problem with Time Windows, where pickups and deliveries can be combined and the ships may in principle never be empty, see [16]. However, in maritime transportation applications, there are often special additional constraints, like for instance ship-port compatibility (can be dependent on the load onboard the ship), ship-product compatibility and stowage constraints. There are often multiple time windows, since many ports are closed for service during nights and weekends and time windows usually span several days.

For an industrial shipping application, the objective is to service all (contract) cargoes while minimizing costs. In tramp shipping, the objective is to maximize profit. All contract cargoes must be serviced, while there may be some optional spot cargoes that may only be carried if there is enough fleet capacity and if they contribute positively to the profit. In both modes, the routing and scheduling will usually be performed by following a rolling

horizon principle, where the plan is updated when new important information becomes available (e.g. new requests for spot cargoes or information about ship delays). For a further description, including mathematical models for different versions of the underlying short-term routing and scheduling problem in industrial and tramp shipping, we refer to [15].

2.2. Strategic planning problems

In this paper we focus on two important classes of strategic decisions for industrial and tramp shipping companies:

- Contract analysis
- Fleet size and mix

Contract analysis includes decisions such as whether to accept or reject a given long-term transport contract and analysis of negotiable terms in the contract. The most common long-term contract in industrial and tramp shipping is the so-called *Contract of Affreightment* (COA), which is a contract to carry specified quantities of a given product between specified ports within a specific time frame for an agreed payment per tonne. In each COA, the shipper or customer will specify cargoes (as described for the short-term routing and scheduling problem) to be lifted by the ship operator. Most often, these cargoes are agreed to be evenly spread throughout the contract period, specifying to some extent how the COA should be divided into cargoes with given quantities and time window widths throughout the year. In such a contract, negotiable terms, in addition to the income rate, might be

- width of the time windows of cargoes,
- flexibility in quantities for each cargo, and
- notice times, i.e. the time in advance the shipper must give a notice about the cargo.

All these terms can substantially influence the possibility the shipping company has to utilize the fleet effectively and improve profit. It is for instance obvious that increasing the width of the time windows can give the ship operator more possibilities to find better routing and scheduling decisions. The same is true when the ship operator is given more flexibility regarding cargo quantity. If the cargo quantity comes in an interval, for instance $10,000 \text{ ton} \pm 10\%$, instead of a single value, there may be more flexibility regarding which ship to use for the cargo. Also if the notice time increases, the planner gets more information earlier that can be utilized to find better solutions.

When a tramp shipping company that also operates in the spot market analyses whether or not a COA will be profitable, it has to make some assumptions about how the spot market will develop for the given contract period. Typically, if the shipping company anticipates low spot rates, it will prefer to have as large contract coverage as possible, or go 'short of tonnage' and vice versa.

Fleet size and mix problems aim at determining an optimal fleet for a given market situation. Only in rare situations is it relevant to determine a completely new fleet size and mix. In most practical industrial and tramp shipping problems, the fleet size and mix problem is more about adjusting or extending an existing fleet. Many shipping companies have fleet renewal programmes that run continuously. Often, the need for adjusting the fleet comes from the scrapping of old ships, while the need for extending the fleet often comes from new COAs or an increased focus on getting a higher share of the spot market.

Contract analysis and fleet size and mix decisions are strongly interrelated. For instance, if a shipping company gets a request for a new major long-term contract it may be necessary to extend or

adjust the existing fleet. The profitability of the contract and the required fleet will also depend on the terms of the new contract (and also the existing ones if it is possible to re-negotiate). It should be noted that whether to accept a contract or not may sometimes also be a question of market share and strategic positioning, and not only a capacity and profit question.

When dealing with strategic planning problems, there is always much uncertainty involved. For instance in a COA, the amount to be carried annually is often given, in addition to information about how the COA should be divided into cargoes with given quantities and time window widths throughout the year. However, this is only loosely specified in the contract, and there is a major uncertainty regarding both the total annual quantities and the times when the transport requests from the COA occur.

In this section we have focused on some of the most common strategic planning problems in maritime transportation. Even though we have focused on fleet size and mix problems and contract analysis, there are also a number of other more specialized strategic planning problems that could be dealt with by the decision support methodology described in Section 3. In Section 4 we present another type of strategic decision that had to be considered for the shipping company in our case study.

2.3. Solution approaches

Even though the above strategic planning decisions are interrelated, it is common to treat them independently to reduce the complexity of the problem. For instance, when deciding whether to accept or reject a given contract request, one analyses the feasibility and profitability for a given fleet. On the contrary, when deciding an optimal fleet size and mix, one usually assumes a given set of contracts or transport requests.

To the authors' knowledge there are only a few contributions in the OR literature on solution approaches for the industrial and tramp strategic planning problems described in Section 2.2, and these are all on fleet size and mix. Actually the first contribution on maritime transportation, by Dantzig and Fulkerson, dates back to 1954 and deals with a fleet size problem in an industrial shipping context [17].

The most commonly used industry approaches for dealing with these problems are various *manual methods*, often supported by simple Excel spreadsheet calculations. Most planners in the shipping companies are practical people with long experience, which they use in such matters, and strategic decisions may often be taken based on their 'gut feeling'. Only rarely they use formalised quantitative decision support models. However, in a few cases the planners also turn to consultancies or researchers that may offer a number of approaches, which can be various *optimization methods* or *simulation methods*.

One type of approach that has been used for solving maritime transportation problems at a strategic level is developing and solving optimization-based strategic models, e.g. mixed integer programming models, see for instance [9]. Some major challenges in this approach are to simplify the problem and to aggregate the data of the problem in a satisfactory manner. To make optimization models solvable for the strategic planning problems considered here, one usually has to omit details on the routing and scheduling aspects of the problem, as well as stochastic elements. Alternatively, the models become very complex and only small instances can be solved to optimality. In addition, such an optimization model most often can be used for solving one specialized problem only. These weaknesses make this approach unsuitable as a general decision support methodology for solving

different strategic planning problems, such as the ones discussed in Section 2.2.

A second type of approach using optimization-based methods is to apply tactical or operational routing and scheduling models and an optimization-based tool for solving the underlying short-term routing and scheduling problem. Then, potential changes of some strategic parameters are specified, based for instance on fleet size or contract negotiations. These form the set of strategic decisions that should be analysed. One often uses historic or forecasted demand data to model scenarios that represent future demand. The scenarios are then entered into this optimization tool and solved one by one, typically for a long planning period (e.g. one year). This planning period is often much longer than for the real short-term routing and scheduling problem. By entering alternative strategic decisions, one can analyse each of these, compare them, and obtain insight to support the relevant decisions. The pioneering work by Dantzig and Fulkerson from 1954 [17] uses a version of this approach, where one actually solves the short-term routing and scheduling problem for various scenarios to provide decision support at the strategic planning level. Only one scenario was specified in [17].

The strength of the latter approach is its ability to handle routing and scheduling aspects, since it relies on a short-term routing and scheduling tool. However, this method will not handle the stochastic aspects of the problem very well. For instance, historic or forecasted demand data may not necessarily give a good representation of future demand. Sometimes there are no historic data, and forecasts may be hard to give. The major drawback with this approach though is that it may provide results that are overfitted due to using complete information: one specifies data for a planning period that is much longer than that which is typical for the short-term routing and scheduling problem. When doing this, an unrealistic amount of information is made available to the optimization-based tool and the solution looks better than what would be achievable in real life, where information is revealed as time goes by.

Simulation is another type of approach that may be applicable to the strategic planning problems considered here. In this method, stochastic aspects of the problem can be treated properly. When simulation is used alone, one usually has to omit or simplify the underlying routing and scheduling decisions. Without taking the routing and scheduling into account, however, the method may fail to correctly identify the effect of the strategic decisions being evaluated.

Analytical methods, which are sometimes an alternative to simulation, would not be applicable in our setting due to the complexity of the strategic decisions involved.

Since optimization often will have shortcomings when it comes to handling stochastic elements, while simulation usually has to omit routing and scheduling aspects, combining these two methods seems to be a promising way. In fact, this has been a growing area of research for the last decades, see for instance [18–22]. In [18], the partnership between the simulation and optimization is discussed, and there is usually one that is subservient to the other. Fig. 1 shows two examples given in [18]. Fig. 1(a) shows the case where an optimization routine is an add-on to the simulation engine, while Fig. 1(b) depicts the situation where Monte Carlo simulation is the add-on used to generate scenarios for mathematical programming formulations. Our methodology, described in Section 3, has elements from both Fig. 1(a) and (b). As in Fig. 1(b), we use Monte Carlo scenario generation, while as in 1(a), we use simulation to evaluate solutions from the overlying optimization problem.

Richetta and Larson combine simulation and optimization in maritime transport system design [23]. Their model has some similarities with our methodology and includes discrete event

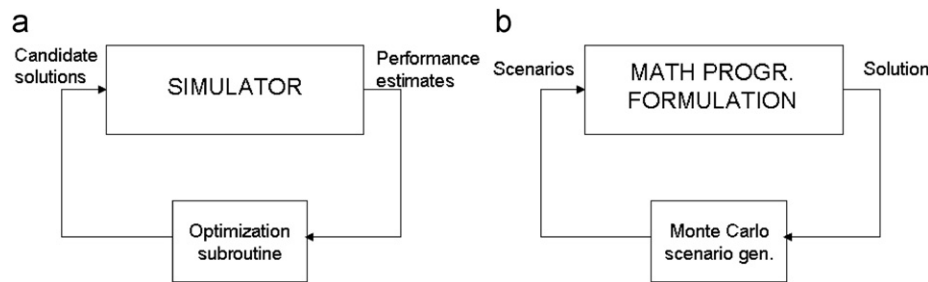


Fig. 1. Optimization for simulation (a) and simulation for optimization (b) [18].

simulation that is combined with dispatch algorithms. The model is applied for fleet sizing and operations planning on New York city's refuse marine transport system. To the authors' knowledge, this is the only paper that combines simulation and optimization that are applied to maritime transport problems that has similarities with the ones dealt with in this paper.

A potential problem with our approach combining optimization and simulation is that the simulation results may show a large degree of variance. That means that the process of obtaining results of sufficient quality may be time consuming, or that it is necessary to include technically demanding variance reduction techniques. There is also a question about obtaining the correct input parameter distributions, although this would apply to other methods as well. Some care is needed in the interplay between the optimization and the simulation: it is important that the optimization reflects the actual short-term planning performed by the shipping company in question, and that the simulation correctly models the underlying stochastic processes so that the optimization is working on realistic data. A potentially major drawback of the method combining optimization and simulation is the technical efforts required, and the fact that few shipping companies have sufficiently qualified personnel to handle these.

3. Decision support methodology

In this section, we propose a decision support methodology for solving strategic planning problems in industrial and tramp shipping, as discussed in Section 2.2. The main idea behind this methodology is to combine the strengths of simulation and optimization to eliminate or reduce the drawbacks with these methods when considered separately. In Section 3.1, we present the conceptual model of the decision support methodology. In the following sections 3.2–3.5, we describe each of the elements of the conceptual model in more detail.

3.1. Conceptual model

The decision support methodology consists of four main processes as illustrated in Fig. 2. We start with **P1**, where a set of *strategic decisions* is determined. Each strategic decision is potentially composed of several individual sub-decisions, such as whether to procure an additional vessel, or whether to accept an additional COA. Essentially, what we term a strategic decision consists of the parameters that the shipping company can control by itself, such as the current fleet, the set of accepted COAs, or the details within a specific COA. On the other hand, there are parameters that are exogenous to the company, and that might be interpreted as stochastic processes. Examples of such processes could range from the movements of weather systems and the breakdown of vessels, to changes in the fuel prices or in the global economy, although the most relevant example would be the processes that control customer demands. In Fig. 2, these are

illustrated by the dotted rectangle on the right hand side. In **P2**, the exogenous information flow is taken as a basis to form what we term *scenarios*. A scenario in our context is a description of a possible future realization of the parameters that are uncontrollable by the shipping company. Although COAs contain many contract terms that are known to the company (e.g., income rate, widths of time windows, cargo quantity windows, and notice times), the exact realization of the cargoes is stochastic: A scenario will then specify actual data for each cargo, such as the exact time of notification and the concrete cargo quantity. In **P3** the strategic decisions and the scenarios are taken as input, and the operational/tactical planning of the shipping company is simulated on each scenario given each strategic decision in turn. This simulation entails solving a *Short-Term Routing and Scheduling Problem* (STRSP) on a rolling horizon basis. Then we perform **P4** based on the results obtained from the short-term routing and scheduling solver to evaluate the effects of the various strategic decisions considered. The dotted arrow from **P4** to **P1** indicates that the analysis of results might allude to new strategic decisions that warrant further investigation. The next four sections elaborate on each of the processes from Fig. 2.

3.2. Determining a set of strategic decisions

The typical initialization step of the methodology is the determination of potential changes of the strategic parameters of the shipping company. These are parameters that influence the daily decisions indirectly, but changing them requires a non-trivial amount of effort on behalf of the company. The previous examples based on fleet size, such as acquiring an additional vessel, or on contract negotiations, such as accepting an additional COA, serve as examples of strategic parameters in our context. Having established the parameters that the shipping company is free to adjust, the decision maker is ready to formulate a set of strategic decisions that should be analysed.

Each strategic decision covers all the aspects that are necessary to guide the operational and tactical planning of the shipping company. For instance, in the contract negotiation example, a description of the strategic decision includes a description of the set of customer contracts that, when taken as a whole decide what the demand for the company will turn out to be. For each contract, there are some key figures that must be determined. The most central ones will be the annual quantities shipped, frequency of cargoes, spread among different loading and unloading ports, typical time windows and information about their flexibility.

Insofar as the strategic decisions influence the contract terms, they are also related to the modelling of the exogenous information processes (cf. the dotted rectangle of Fig. 2). Not only should the information explicitly stated in the contracts be involved, but also the historical statistics and the experience of the planners at the shipping company. Because of the high degree

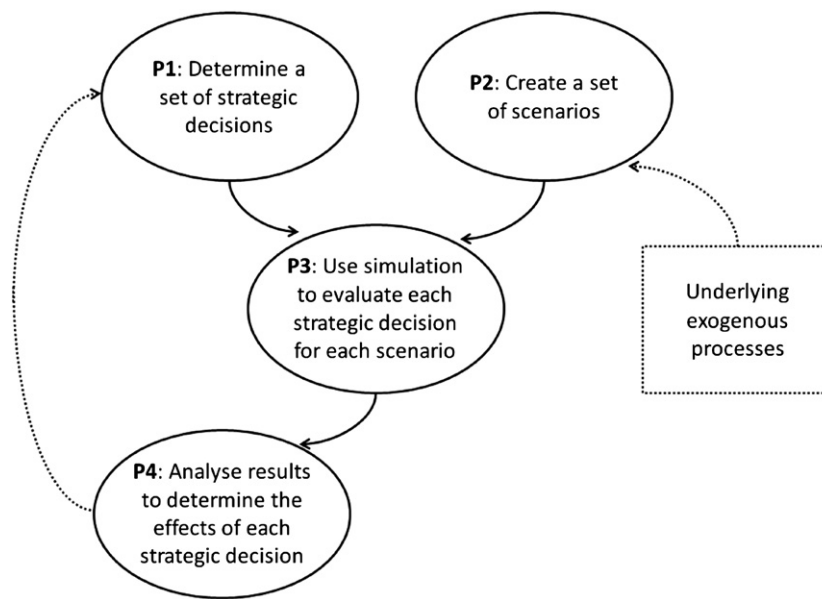


Fig. 2. Conceptual model for decision support methodology.

of uncertainty related to the information about future demand, probability distributions for different outcomes must be decided. Some customers have clear demand patterns, while others have a higher degree of randomness. The way that the demand is distributed over the year must thus be considered for each contract separately.

For shipping companies with a substantial share of spot cargoes, the spot market must also be modelled in a best possible way. This market is normally characterized by a high degree of randomness, but our methodology based on Monte Carlo simulation would be appropriate to handle this property.

Another important strategic parameter is the specification of the fleet, such as the number of vessels and more detailed description of each vessel. In business cases where the availability of the vessels is an uncertain factor, probability distributions to use in the Monte Carlo drawing should also be entered for the fleet. However, the availability of the fleet is often predictable, and in the remaining sections of this paper we will consider the fleet as given.

The determination of a particular set of strategic decisions can sometimes influence the stochastic processes that are used when creating scenarios in **P2**. For example, if the strategic decision involves a change in the terms of a COA, such as quantities to be shipped in each cargo, this might in principle affect the processes that result in a request by the customer. If this is the case, one should generate scenarios (in **P2**) that are specific to each strategic decision, and only perform simulation (in **P3**) for those combinations of strategic decisions and scenarios that are consistent.

3.3. Scenario generation

When the strategic decisions have been determined, the next step is to generate scenarios (**P2** in Fig. 2). The scenarios are based on the assumptions made about the processes of which the shipping company has no control, such as the concrete build-up of demand by the customers or the effect on travel times/costs induced by bad weather. The generation of scenarios should only involve processing the related probability distributions, and making calculations based on these. This is achieved by making use of Monte Carlo sampling from the probability distributions.

Rubinstein and Kroese [24] give an in-depth presentation of the Monte Carlo method.

This way of generating demand scenarios to use in our analyses, makes it easy for us to make as many scenarios as we want for each strategic decision with little computational effort. The scenarios are sent one by one through the solving process (**P3**), until all of them have been processed for each strategic decision. With a higher number of scenarios, the uncertainty of the results is decreased according to normal statistical rules.

The most important part in our definition of a scenario is the complete set of cargoes that the shipping company is requested to transport during the scenario period, in other words, the demand.

3.4. Evaluation through simulation

This process concerning solving the short-term routing and scheduling problem (STRSP) (**P3**) involves working with each scenario, acting like the planner at the shipping company. The planner's task is to find the best possible routes and schedules for the given demand.

For this task, a solver for the underlying STRSP is needed that is based on OR methods. For the type of analyses we are interested in, the focus should be on recreating the real planning situation rather than finding the optimal solution to each short-term planning problem. When also considering the computational complexity of solving the STRSP, a heuristic solution method seems to be most appropriate. An increasing number of shipping companies are shifting from relying on purely manual planning methods to using optimization based decision support tools when making short-term plans involving routing and scheduling of their fleet.

The demand for the full scenario period should be revealed to the solver gradually to make our model resemble the real planning situation in the best possible way. Since the scheduling horizon for a shipping company typically is much shorter than the scenario length, revealing the whole scenario at once would lead to unrealistically good scheduling decisions (as discussed in Section 2.3). Thus, we want our model to work its way through the scenario period iteratively, using a rolling scheduling horizon, as shown in Fig. 3.

The length of the scheduling horizon, H , should be set according to the horizon actually experienced by the real planner, and the solver is only presented with the demand within this horizon. The scheduling horizon, H , will usually be derived from the nomination agreement of the COAs (i.e. cargo notice times), since there is usually no point of planning further ahead in time than cargo information is available. When the routing decisions for this horizon have been made, the methodology will make a jump in time of proper length ($K \leq H$). The present time (t) in the simulation is updated and new demand information is revealed at the end of the scheduling horizon. Updates of the positions of the vessels will be made using the latest planned routes. What was planned up to the new present time is considered executed. With the new demand information and starting positions for the vessels, the solver can recalculate the routing decisions and make a new plan.

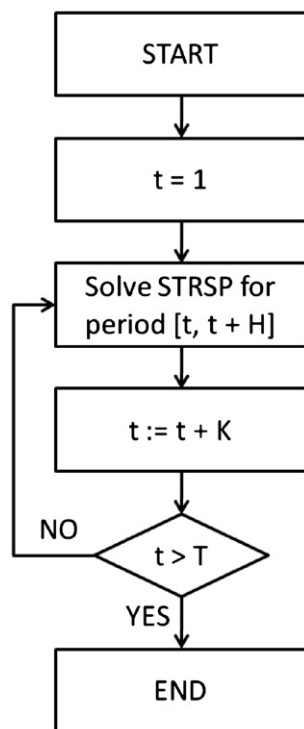


Fig. 3. Simulation framework used in P3.

Fig. 4 illustrates the shifting of the scheduling horizon between two random iterations. Here we have used $K=3$ days as the length of the jump in time simulated between iterations, and it is assumed that there only exists certain demand information for the next 7 days (H).

3.5. Analysis of results

For every scenario, the solving process of the STRSP will provide some important output figures. The economic results in terms of net daily income, total income, or total costs are naturally the most important results. In cases where feasible solutions to the problem cannot be found without exceeding the time windows, other outputs can have relevance for the evaluation of the solutions. Interesting figures could then be the number of cargoes not served or the total delay. These kinds of situations also happen in reality and forces the planner to break the agreement with some customers, risking a loss of goodwill, or to buy extra capacity from the spot market.

When several generated scenarios are run through the solving process, the average results for each strategic decision must be calculated. As mentioned earlier, a higher number of scenarios would lead to a lower standard error in the results. The standard deviation of the results will be an important measure of the uncertainty involved.

After gathering the results from the simulations, these must be interpreted to give an evaluation of the strategic decisions. The objective is to identify any changes in the strategic parameters that seem beneficial according to the interpretation of the results. Low capacity utilization, for example, might indicate that the fleet could be reduced.

If the effects of specific strategic decisions are considered, the process can be run for all of the alternatives. Afterwards, the results can be compared, and the best decision can be made. If the decision under consideration does not directly involve the customers, the same scenarios can be used for each of the strategic decisions, to make comparisons as efficiently as possible, through matched pairs testing of the difference in performance. This is for example the case when evaluating fleet size or the length of the scheduling horizon (i.e. notice times in COAs). However, as noted in Section 3.2, it can happen that scenarios specific to each strategic decision are required, in which case pair wise tests are not meaningful.

As long as the analysis indicates that favourable changes in the strategic parameters are left untested, a loop back to the start of the methodology should be followed. The strategic parameters should be set to seemingly better values and the processes re-run.

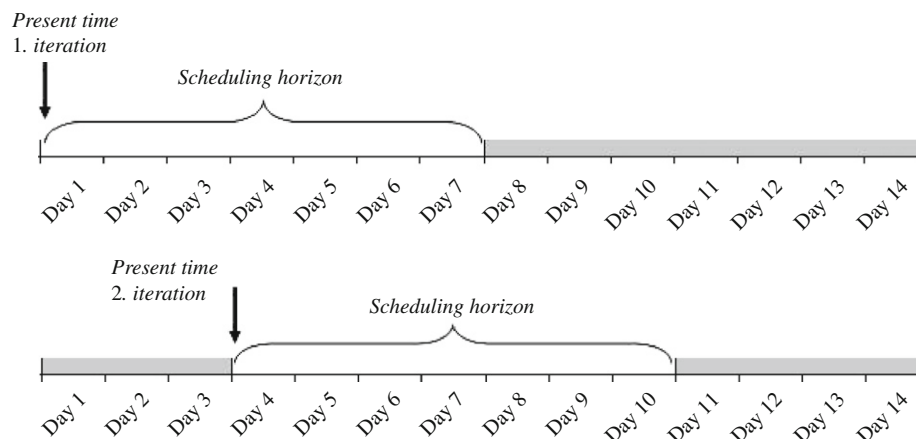


Fig. 4. Rolling scheduling horizon principle.

4. Case study: Wilson—the Norwegian Rhine line

An implementation of the methodology has been applied and tested on a case study for the Norwegian shipping company Wilson ASA. In Section 4.1 we describe the software used in the implementation. Section 4.2 contains a description of the case and isolates the strategic decisions that will be examined. The details around the scenario generation process is found in Section 4.3, while Section 4.4 summarizes the results.

4.1. Implementation of decision support methodology

The methodology has been implemented using TurboRouter as the tool for solving the underlying routing and scheduling problem. TurboRouter is a decision support system designed for ship routing and scheduling in industrial and tramp shipping [4]. The system uses an efficient multi-start local search heuristic as a solver for these problems [3]. MS Excel is used for input of case and scenario information to TurboRouter and for output of results from the analysis.

The scenario building is carried out by entering some static data such as port locations and distances directly in TurboRouter. The dynamic data that can differ between scenarios are stored in Excel sheets. This is typically fleet and demand information. Macros made in Visual Basic perform the scenario generation process by transforming these data into detailed demand instances of desired length, and store them in text files suited for TurboRouter.

The original version of TurboRouter has been configured to implement the rolling scheduling horizon. For each instance, TurboRouter gives output on sailing schedules and economical figures, which are used to analyse and evaluate the scenario results.

4.2. Case study description

The Norwegian Rhine Line (NRL) is a fleet segment of Wilson operating in the North Sea area, see Fig. 5. It has small vessels that

are able to operate on the river Rhine as well as across seas. The fleet consists of 10 vessels with capacities ranging from 1500 to 2400 deadweight tons.

Wilson has two different categories of customers on their NRL. For the north-going transportation, the Norwegian steel industry is the important customer category, importing a large annual quantity of steel products from the Ruhr area in Germany. The main loading port for the steel products is Duisburg, located 100 nautical miles up the river Rhine. Capacity limitations for the vessels are determined by the water level in the river, and barges are often needed to carry some of the steel to Rotterdam, where the loading is completed. The steel is delivered to a large number of ports along the Norwegian coast with fluctuating frequency. A vessel typically unloads in 3–4 ports on each voyage.

The other main customer category is the Norwegian production industry exporting bulk products to the European market, mainly to Germany, the Netherlands and the UK. These customers have strict time windows for cargo pick up.

Both categories of customers have long-term contracts with Wilson, and thus the NRL only operates in the spot market on rare occasions. Approximately 720 cargoes are lifted annually. The average cargo size is around 1000 ton, meaning that one often has to combine several cargoes onboard the ships. Even though the COAs are long term, the nomination agreements only give a cargo notice time of one week. Since this is more or less the same for all COAs, the NRL planner experiences a short planning horizon of only one week. The planner allocates cargoes to vessels manually, focusing on avoiding waiting time due to ports that are closed at weekends.

Two important strategic parameters for NRL were addressed in the case study:

1. Should Wilson-NRL continue sailing the Rhine or should they only sail to Rotterdam and hire barges from there?
2. Contract negotiations: What is the effect of renegotiating notice times from 1 to 2 weeks?

The first is a case-specific decision, while the latter is a typical contract analysis decision, as discussed in Section 2.

4.3. Details of the case study

From the strategic parameters to be addressed, four different decisions were derived:

R1. With Rhine-traffic and one week planning horizon (notice time).

R2. With Rhine-traffic and two week planning horizon (notice time).

NR1. Without Rhine-traffic and one week planning horizon (notice time).

NR2. Without Rhine-traffic and two week planning horizon (notice time).

For the set of four strategic decisions, 12 different scenarios with a length of 12 months each were generated. The full-year length of the scenario period makes it easier to cope with seasonal demand variations, and reduces the impact of the start- and end-effects of the simulation. With such a long scenario period, most of the statistical variance is partly evened out within each scenario, and fewer scenarios are needed to obtain statistically significant conclusions.

The input sheets for the different contracts were tailored to fit their statistical characteristics. Table 1 shows an example of a sheet for determining the number of cargoes nominated by one particular customer for a given week. For each week a new random variable is drawn and the statistical distribution is used

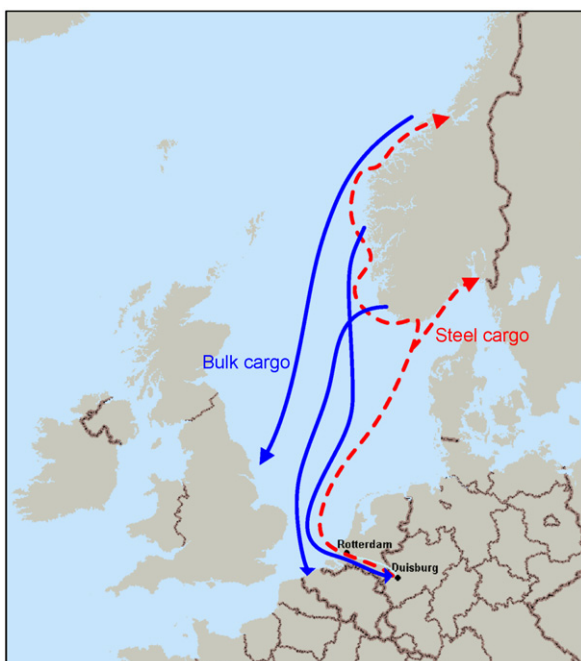


Fig. 5. Operation area.

Table 1

Example of generating the weekly number of cargoes from a COA.

Number of cargoes per week	Distribution	Accumulated	Lookup value
0	17%	–	0
1	48%	17%	1
2	33%	65%	2
3	2%	98%	3
4	0%	100%	4
Control should be 100%	100.000%		
Stochastic variable to control output	0.440167541		
Included seasonal variations	0.440167541		
Number of cargoes per week	1		

Table 2

Example on deciding quantity of a given cargo using a beta distribution.

Quantity	Distribution parameters	
Lower limit	600	
Upper limit	2300	
Alpha	2.000	
Beta	3.000	
Expected value	1280.000	
		Random variable
Quantity for first load	1476.7	0.710733797
Quantity for second load	1548.3	0.768938675
Quantity for third load	1292.1	0.537114334

to determine the number of cargoes nominated this week. The statistical distribution used is based on historical data, and takes into consideration seasonal variations. For the week illustrated in this example the number of cargoes nominated is one. For other customers, the number of cargoes is smaller, and instead of generating the number of cargoes nominated per week we generate the number of weeks until the next cargo is nominated. In total there are nine separate processes for generating cargo numbers, four of which are of the latter type. For each process, a suitable distribution and parameters are determined based on historical data available and knowledge from the company's planner.

In addition to variation in the number of cargoes, the size of each cargo is also uncertain and must be drawn based on probability distributions. The type of distribution should be tailored according to the statistical characteristics of each customer. For most customers we use, as illustrated in Table 2, a beta distribution [19] to model the cargo sizes. Lower and upper limits can be set, and the alpha and beta values are set to shift the expected value of the distribution. Here, the cargoes can be from 600 to 2300 ton, while the probability of drawing a quantity around the expected value of 1280 ton is large. For some of the customers the cargo sizes are more naturally modelled using discrete distributions, especially for cargoes that require a full ship load. The cargo size may also be subject to seasonal variation, which means that the distribution parameters depend on the time of the year.

The cargoes have specified loading ports and discharging ports. For some customers one or both of these may be fixed, whereas for others they are generated from general discrete distributions. There are also time windows associated with each cargo, however, in this case they are very simple and do not include any stochastic component.

Furthermore, the scenarios specify water levels for the Rhine, as these have consequences for the transportation on the river as well as for the cost of hiring extra barges when levels are too low for the ships in Wilson's fleet. The water level of the Rhine varies

quite a lot from day to day or month to month, and an AR-1 type autoregressive model [25] is used to simulate the water level.

The scenarios generated varied from 695 to 739 cargoes with about 750,000 ton to be carried in total. Although the number of cargoes is a result of observing hundreds of random variables, as outlined above, the total number of cargoes remains quite stable. Each cargo also depends on several random variables to determine attributes such as quantity and loading and unloading ports. This means that the number of input parameters to the simulation model is quite high. Each of the input parameters, as well as the choice of distribution to use, was determined from historical data and through discussions with the shipping company in a time consuming process. The accuracy of a simulation should always be questioned. In this case, the scenarios were produced after close cooperation with the NRL planner in Wilson, who examined and finally approved them as being realistic.

The scenarios have been generated in a very straightforward process, without using any involved techniques, such as variance reduction techniques [24]. However, as discussed in Section 3.5, in this case the strategic parameters do not influence the scenario generation. Hence we are able to use the technique of *common random numbers* to reduce variance in the final computational results when comparing alternative decisions. In other settings more elaborate techniques may be required, although this is outside the scope of this paper.

For the rolling scheduling horizon, jumps of $K=3$ days in time were used since this is the average time between each re-planning made by the planner at Wilson. This implies that for a scenario period of one year TurboRouter must for each scenario make more than 120 recalculations. The problem size for each recalculation depends on the scheduling horizon length (i.e. one or two weeks). The computation time for each scenario was approximately 24 h when applying an Intel Pentium M 1.83 GHz, 1 GB RAM. It is clear that a manual procedure based on the same level of detail would be too time consuming to use as a basis for evaluating the strategic decisions.

4.4. Case study results

The results showed that the two strategic parameters that Wilson was interested in are relatively independent of each other, one decision giving the same effect regardless of how the other decision is made. Thus, the two decisions will be discussed separately. Table 3 presents selected key numbers for each of the four strategic decisions, labelled R1 (with Rhine traffic and notice times of one week), R2 (with Rhine traffic and notice times of two weeks), NR1 (without Rhine traffic and notice times of one week), and NR2 (without Rhine traffic and notice times of two weeks). Table 4 presents results for pair-wise comparisons between selected decisions (R1 vs. R2, R1 vs. NR1, and NR1 vs. NR2), where lower and upper bounds for 95% confidence intervals for

Table 3

Average key numbers for each of four strategic decisions, after simulation on 12 different scenarios.

	Voyage costs	Barge costs	Total costs	Cargoes lifted
R1	46,418,058	3,000,659	49,418,717	694.8
R2	46,238,552	2,975,946	49,214,498	698.6
NR1	42,854,849	–	42,854,849	698.9
NR2	42,624,026	–	42,624,026	700.5

Table 4

Differences in total costs, using pair-wise comparisons between selected strategic decisions, based on 12 scenarios.

	Avg	StD	LB	UB
R1 vs. R2	204,219	481,656	–101,813	510,251
R1 vs. NR1	6,563,868	627,329	6,165,279	6,962,456
NR1 vs. NR2	230,823	407,407	–28,033	489,679

Table 5

Differences in total number of cargoes lifted, using pair-wise comparisons between selected strategic decisions, based on 12 scenarios.

	Avg	StD	LB	UB
R1 vs. R2	–3.8	2.8	–5.6	–2.1
R1 vs. NR1	–4.2	2.2	–5.6	–2.8
NR1 vs. NR2	–1.6	2.4	–3.1	–0.1

the difference in total costs are shown in columns LB and UB. Table 3 similarly presents confidence intervals for the difference in the total number of cargoes lifted for the same pairs of strategic decisions. Notice that since all strategic decisions are evaluated on the same set of scenarios in this case, the confidence intervals can be calculated based on matched pairs to reduce the variance.

The computations showed that the effect of increasing the notice times of the contracts, and hence the planning horizon, from 1 to 2 weeks was insignificant in terms of total costs as presented in Table 4. It was therefore not recommended that Wilson should risk any reduction in prices or loss in goodwill by forcing increased notice agreements onto their customers. However, as is presented in Table 5, there is an improvement with respect to the number of cargoes lifted when the notice times are increased to two weeks. On average, about 20 out of 720 cargoes are missed when the fleet is limited to 10 ships. In practice, this would be solved either by violating the time windows, e.g., serving the transportation request one week too late, or by hiring extra capacity for a short time. In any case, the small, but significant, reduction of missed cargoes by on average 3.8 would not be worth the trouble of renegotiating the contract terms of all of Wilsons customers.

The total costs are reduced by about 13% when not sailing up the Rhine. This is a significant cost saving (see Table 3) that should be compared to the costs that will arise when moving cargoes from and to Rotterdam. However, this trade-off was something Wilson would evaluate and was not a part of this study, which only provided the input regarding the above cost effects. In addition, when not sailing the Rhine, there is also a significant increase in the number of cargoes carried. Again, this change is rather small, but as can be seen in Table 5, if one chooses to avoid sailing the Rhine the further increase in cargoes carried that can be achieved by increasing the notice times is reduced.

The results from our methodology provided important decision support and new insights regarding the above strategic decisions. For instance, it was quite surprising that the effect of increasing the notice times from 1 to 2 weeks was so small. One possible explanation is that one is not able to utilize this because the sailing times are a good match to the weekly cycle, with the ports being closed for service during weekends. This combined with the fact that the cargo time windows are narrow probably makes it hard to utilize the extended notice time (and planning horizon). Furthermore, the capacity of the fleet is already utilized almost to the full, and there are no spot cargoes that could be serviced to increase the income.

The results were also evaluated by the planners in Wilson. In addition, they evaluated the input parameters (discussed in Section 4) and the methodology used, and had confidence in the results. This of course is very important when making these important strategic decisions. In this case it would be prohibitively time consuming to evaluate the decisions and scenarios using manual techniques, and it is not possible to do an analytical evaluation of the complex underlying transportation system.

5. Concluding remarks

In this paper, we have presented a new decision support methodology for strategic planning in industrial and tramp shipping. Our proposed methodology combines simulation and optimization, where a Monte Carlo simulation framework is built around an optimization-based decision support system for short-term routing and scheduling. We have applied a rolling horizon principle in the optimization, where information is revealed as time goes by. There are two major advantages with this solution approach compared with other approaches:

1. It makes it possible to deal with the stochastic aspects, while at the same time considering the underlying routing and scheduling aspects of the problem.
2. It is flexible in the sense that it can easily be configured to provide decision support for a wide range of strategic planning problems, such as fleet size and mix problems, analysis of long-term contracts and contract terms.

One main disadvantage with the methodology is that the analysis requires a lot of computer time due to the many runs that must be made to provide trustworthy results. Another drawback is that relatively large amounts of high quality data are required to generate probability distributions for demand and other uncertain parameters. However, we believe that computer time is not critical when analysing important strategic decisions that may have an impact several years ahead in time. We also believe that in many industrial and tramp shipping problems, one can use historical data combined with the planners' experience to generate probability distributions that reflect reality close enough to give valuable support. To execute the analysis, with combined optimization and simulation, highly qualified personnel may be required. Although few shipping companies have such personnel available, it may be worthwhile to hire consultants to perform such analysis when evaluating strategic decisions with high impact.

We have tested the methodology on a real case for a major Norwegian shipping company. The methodology proved to provide valuable decision support on important strategic planning problems for the shipping company. In this real case, the shipping company has almost 100% contract coverage, i.e. almost all cargoes carried come from long-term contracts (COAs). We therefore only had to model each of these contracts (as described

in Sections 3.2 and 4.3), and we did not have to consider modelling the spot market. For tramp shipping companies also operating in the spot market, we would need to model this in an appropriate way, which may not be trivial. This is an interesting topic for future research.

Another interesting area for future work is how to analyse the effects of each potential strategic decision (**P4** in Fig. 2). At present, the evaluation and comparisons of alternative decisions are made by manual inspection. This could ideally be done by an automatic process that will also guide the generation of new potential strategic decision to be evaluated in a direction that seems to be promising based on the previous results.

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